University of Stuttgart
Institute of Industrial Automation and Software Engineering

Image Augmentation and Transformation Techniques for Enhanced Scene Comprehension in Vision-Language Models

FA 3985

Submitted at the University of Stuttgart by
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Masters in Electrical Engineering

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20.08.2026

Abstract

Deploying Vision-Language Models (VLMs) on autonomous mobile robots is severely constrained by the strict computational, memory, and thermal limits of edge hardware. While fine-tuning or retraining foundation models for adverse operational environments is impractical on embedded platforms, compact edge decoders frequently suffer from spatial grounding deficits, resolution downsampling loss, and temporal redundancy in continuous video streams. This thesis investigates an inference-time visual preprocessing architecture designed to enhance VLM scene comprehension and downstream reasoning without modifying underlying model weights.

We propose and evaluate four task-aware spatial prompting techniques Context-Aware Zoom (CAZ), Query-Aware Bounding Box (QABB), Semantic Hazard Isolation (SHI), and Perception Metric Grounding (PMG) alongside dual-track query-aware video keyframe extraction pipelines (QA-IF and Smart Sampling). To systematically benchmark these strategies across the cognitive hierarchy of mobile robotics, we curated a benchmark corpus spanning seven functional task domains, comprising 2,103 static QA pairs across 731 visual scenes (with a 30% low-light stress test) and 70 continuous robotic video trajectories annotated with minimal keyframe triggers. Real-time tensor inference and telemetry profiling were benchmarked on an NVIDIA Jetson AGX Thor across three representative model tiers: LLaVA-Phi3 (3.8B), Qwen3-VL (4B), and Gemma-4 (12B). To overcome the blind spots of conventional lexical metrics, deep semantic correctness was evaluated via an automated, three-round heterogeneous multi-agent debate jury (DeepSeek-R1, Qwen2.5, and Llama-3) offloaded to an HPC cluster.

Across 12,250 multimodal benchmark instances, our empirical findings reveal that inference-time adaptation fundamentally redefines edge vision-language perception through two governing trade-offs: attentional focus versus global context in static imagery, and extraction latency versus token processing economy in continuous video. While targeted spatial prompts like bounding overlays and zoom crops improve fine-grained object localization by up to +8.81%, they impose up to a -19.53% penalty on open-ended scene understanding by stripping peripheral context—demonstrating that visual prompts must be applied conditionally rather than globally. Under adverse sensing conditions, lightweight photometric regularizers such as Gamma Correction reliably stabilize low-light hazard detection without model retraining. In dynamic video streams, while traditional shot detectors fail catastrophically (94.29% error rate) under smooth robot camera motion, native container demuxing achieves zero-drop real-time throughput (0.45 s latency), and query-conditioned frame pruning compresses visual payloads down to 1.74 frames to prevent downstream edge VLM token saturation. Ultimately, our analysis uncovers a structural decoupling where standard NLP metrics reward linguistic fluency while overlooking physical hallucinations, establishing that task-conditioned visual preprocessing paired with grounded multi-agent factual auditing provides a practical, robust foundation for embodied edge autonomy.

die Zusammenfassung

Schreiben Sie eine Zusammenfassung Ihrer Arbeit und nennen Sie hier Ihre wichtigsten Ergebnisse.

Die Zusammenfassung sollte sowohl auf Englisch als auch in einer zweiten Sprache verfasst werden, falls dies möglich ist, unabhängig davon, ob die Arbeit auf Englisch oder in der von Ihnen bevorzugten Sprache verfasst wurde. Diese sollten auf separaten Seiten stehen, die englische Version zuerst.

Preface

This Master's thesis was conducted at the Institute of Industrial Automation and Software Engineering (IAS) at the University of Stuttgart. The research presented in this work investigates the intersection of embodied artificial intelligence, edge computing, and robotics perception, focusing on how inference-time visual prompt engineering and dynamic video keyframe selection can enhance scene comprehension in Vision-Language Models (VLMs) deployed on mobile robotics platforms.

The primary motivation behind this work stems from a practical challenge in autonomous systems: while multimodal foundation models exhibit exceptional reasoning capabilities, their deployment on resource and energy-constrained edge devices remains bound by strict latency, memory, and thermal limits. By exploring non-invasive visual transformations and container-level video extraction pipelines, this research aims to bridge the gap between heavy cloud-level reasoning architectures and real-time onboard robotic control.

Acknowledgments

First and foremost, I would like to express my sincere gratitude to **Prof. Dr.-Ing. Andrey Morozov** for granting me the opportunity to conduct this research at the Institute of Industrial Automation and Software Engineering (IAS) and for providing the academic vision, resources, and environment essential for completing this work.

I am deeply grateful to my supervisor, **Joachim Grimstad**, for his continuous guidance, insightful discussions, and dedicated support throughout this project. His constructive feedback, technical expertise, and encouraging advice were instrumental in refining the methodology, experimental rigor, and presentation of this thesis.

I also extend my appreciation to the researchers, colleagues, and technical staff at the University of Stuttgart for maintaining the high-performance computing clusters and edge robotics hardware platforms that made the extensive empirical benchmarks and multi-agent evaluations possible.

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Date and Signature: